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A New VC Dimension Based on Probability

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Abstract. This paper proposes a new definition of VC dimension based on probability. The VC dimension is the most important parameter to evaluate the learning ability of a hypothesis set. However, in many actual problems, the origin VC dimension is always too large to use. And there always exists a paradox between the size of hypothesis set and the range of confidence interval. This paper solves these problems by taking the information of data set and probability into consideration. We propose a new definition of VC dimension based on probability and gives a stronger VC bound.

Introduction

The three core factors of machine learning problems are the size of hypothesis space, the size of training data set and the range of confidence interval. In general, comparing to the other two factors, training sample problems are easier to solve. Consequently researches on machine learning majorly focus on the hypothesis space and confidence interval. Among them, the probably approximately correct (PAC) is one of significant machine learning theories and there are plenty researches on it [7], [8], [9]. The VC dimension theory proposed by V.N.Vapnik [4] is the key to understand the PAC theory. In [4], Vapnik developed an inequality relationship between the size of hypothesis space and the range of confidence interval. In addition, the theory has become the most classical one in machine learning with many related researches [1],[2],[3],[5] on it.

The VC dimension [4] is defined by considering all training samples on any possible distributions. However, one fact has been neglected that a machine learning algorithm in a real problem is not necessary to consider all distributions. In other words, if only most cases are considered, the algorithm can be possible to obtain high performance.

Usually, a paradox between the training error and the range of confidence interval will always exist. On the one hand, the more complex the hypothesis space is, the smaller training error we get when learning. On the other hand, according to the inequality of Vapnik [4], the larger the hypothesis space, the bigger the confidence interval is. Vapnik has put up an inequality to explain this paradox and introduced VC bound [12]. The VC bound is an important parameter of a classifier and always useful when designing a learning machine [6], [10], [11].

In this paper, we will define a new VC dimension based on probability. We take the information of data set into consideration. Information of data set can always be gathered in real problems. By using this information, new VC dimension and VC bound are introduced. They offer a theory that we can classify bigger sample set by using the same hypothesis set without the loss of accuracy. This means we can find a balance between the complexity of hypothesis space and the range of confidence interval.

Basic Definition

The Original Definition of VC Dimension

The proposal of VC dimension is intended to research on the speed of uniform convergence in learning progress. It is an important parameter based on statistic theory to measure the learning ability of the hypothesis space.

Definition 1: For hypothesis space S, if there are N samples which can be classified in allkinds by hypotheses in the space S, we claim that the hypothesis space can shatter N samples.

Definition 2: The VC dimension of the hypothesis set S is defined as the maximum number N of sample which the hypothesis space can shatter. If the hypothesis can shatter any size of samples, then the VC dimension is infinite.

As the core part of the modern theory of machine learning, VC dimension is the most important parameter to measure the learning ability of hypothesis space. The bigger the VC dimension, the more complex problem the hypothesis space can handle.

The original VC dimension of Vapnik suits for any problems with arbitrary distribution of sample. That is why the VC dimension is universally applied, but it cannot get the satisfying result of many learning problems in practical. The VC dimension could always be too big to lose its actual use or even lead to over-fitting when there are noises in the training samples.

Approaches to Define VC Dimension on Probability

Based on the consideration to broaden the definition of VC dimension, we propose new concepts of VC dimension in three aspects:

(1).consider the influence of distribution of samples on the VC dimension.

(2).consider the situation where training error could not be zero.

(3).add probability into discussion.

First introduce the following notations:

Definition3: S is the hypothesis space. If the classification of every sample only has two consequences, we call this classification dichotomy.

Definition 4: S is the hypothesis space. The maximum of dichotomy which S can produce with any N samples is defined as growth function $m_s(N)$.

Definition 5: h is one hypothesis of the hypothesis space S. Define $E_{in}(h)$ as its training error.

The particular formula to calculate training error should be confirmed by the designer of the learning machine.

In this paper, we suppose it to be

$$E_{in}(h) = \sum_{i=1}^{n} (h(x_i) - y_i)^2 \tag{1}$$

In the error function, y_i is the value of given function of the sample x_i, $h(x_i)$ is the calculated value of the sample x_i according to the chosen hypothesis function h .

This function shows the degree of the deviation of the chosen hypothesis towards target function.

Probability VC-dimension(PAC)

We suppose that the original VC dimension of the hypothesis S is k, and the amount of training samples is N. In order to simplified problem, we can just consider it as one time dichotomy problem generally.

Consider the situation where training error could not be zero

In classical definition of VC dimension, the training error is always supposed to be zero. Not only could this constraint makes the hypothesis space so large that aggravate the VC dimension in great deal, but also can procedure easily incite over-fitting when there exists noise in samples.

Consequently, it will make the function from hypothesis space work perfectly for the training data, but not quite suitable for the real data space. So, we consider the new definition of VC dimension under the condition that the training error could not be zero.

First we introduce new definition of training error.

Definition6: regarding the given constant η, $p(\eta)$ is the probability of that the inequality $E_{in}(h) \le \eta$. In another word, if we fix the probability $p(\eta)$ of every η, we define the new definition of training error as $E_{in}(h)$ and the probability of that $E_{in}(h)$ can fulfill the inequality $E_{in}(h) \le \eta$ is $p(\eta)$.

Based on the new definition of training error, we can do pruning on hypothesis space. There are two ways to do so: the first method is to delete hypotheses from the original spaces directly. The principle of this method is: if two hypotheses h_1 and h_2 from set S meet the requirement that they can make the same classification of any sample from data set under the definition of the new training error, then we can regard them as the same hypothesis. We should delete hypothesis under principle to acquire a new hypothesis space where there is only one sort of hypothesis.

The other method is that we curtail the $m_s(N)$ dichotomies that are produced by the original set S . The procedure of dichotomy from hypothesis S is a sort of classifying process. And the principle to delete dichotomies is to delete the same classification under the definition of new training error.

The two methods can both delete hypotheses in the set and then diminish the hypothesis space and shrink the VC dimension. Though the principle of the two pruning methods is the same, the first way is to diminish the hypothesis set according to the given sample ,and the second way is to deal with the samples that will generate the maximum number of dichotomies(because there are $m_s(N)$ dichotomies).

If we do not consider the influence of the distribution of the data sets on VC dimension, we should adjust the second pruning method because a new VC dimension could be found which behaves well in any data distribution. Otherwise we should use the first method. Both methods can diminish the hypothesis to shrink the space. The effect on the VC dimension of the data distribution will be discussed in the following.

Consider the Influence of Distribution of Sample

The original VC dimension is defined on all possible data sets by Vapnik, which means no constraint on samples. However, in most applications, samples always have their patterns. In a specific distribution, we can classify N samples correctly using a small hypothesis set with a small VC dimension.

In practice, we can get rid of many hypotheses when the data obey a certain distribution. It means we can significantly shorten the hypothesis set and diminish the VC dimension while the output of the learning is not affected.

Definition7: S is the hypothesis set. For any N samples which follow the given distribution F, we define the maximum number of dichotomies that the hypothesis produces as $m_s(N)$.

Definition8: S is the hypothesis set, k is the original VC dimension of S. For any N samples, the amount of dichotomies is A. The probability that S classifies them correctly is

$$P = \frac{A}{2^N} (P \le P_0, P_0 = \frac{m_s(N)}{2^N}) \tag{2}$$

So we can define the VC dimension of hypothesis set S is N with probability P .

The first improved plan is as follows. In an actual problem, the amount of hypotheses is N. Suppose K in the whole 2^N kinds of classifications will never happen. Then for a hypothesis of which the VC dimension is k, the probability of classifying correctly is

$$P = \frac{A}{2^N - K} \tag{3}$$

If K is large enough, the probability above can be close to one.

However, the improvement above is still too idealized because it is hard to determine that which classifications can never exist. So we introduce probability into improvement and come up with the new plan below.

In an practical situation, the probability of each element in all 2^N classifications is $P_i, i \in \{1,2,...,2^N\}$. Let us just suppose that each classification is independent to another, so we have

$$\sum_{i=1}^{2^N} p_i = 1 \tag{4}$$

For a hypothesis whose original VC dimension is k ,the probability of classifying correctly is

$$P_S = \sum_{i=1}^{2^N} p(s|i) \times p_i \tag{5}$$

$p(s|i)$ is the probability of learning correctly in the condition when s belongs to i classification. Since the problem it learns is a dichotomy problem, if the i_{th} classification is included in the dichotomies, then $p(s|i) = 1$, else, $p(s|i) = 0$.

Thus,

$$P_S = \sum_{i=1}^{2^N} p(s|i) \times p_i = \sum_{\substack{i=1 \\ i \in A}}^{2^N} p_i \tag{6}$$

According to the formula above, if we ignore the difference of the probability of each sample or say that the probability of every sample is the same, then we can get

$$p_i = \frac{1}{2^N} \tag{7}$$

Thus,

$$P_S = \sum_{i=1}^{2^N} p_i = \frac{A}{2^N} \tag{8}$$

This is the formula in the past chapter. This explains that the new definition is expanding of the original one.

However, since the probability of every sample is hard to measure accurately and there will be some classifications with a tiny probability, there is no need to enlarge the hypothesis set. Then we have the rules below.

Definition9: For a certain learning practice, if the probability of each element in all 2^N classifications is

$$P_i, i \in \{1,2,...,2^N\} \tag{9}$$

We set a threshold t, if $p_i < t$, then we cancel the classification in the set. We also delete the hypotheses in hypothesis set. If $p_i \geq t$, we keep them in the two sets. The new hypothesis set after pruning is called $S^{'}$.

Suppose that we delete K classifications and the amount of hypotheses of $S^{'}$ after deletion is $m^{'}_{S^{'}}(N)$. According to the definition, in the curtailed hypothesis set, the ones that $p_i < t$ will not exist.

So the probability of correct classification is

$$P_S = \sum_{\substack{i=1 \\ p_i \geq t}}^{2^N} p(s|i) \times p_i \tag{10}$$

If the classification set $S^{'}$ has the i_{th} classification, $p(s|i) = 1$, else, $p(s|i) = 0$.

Definition of PVC Dimension

We formally introduce the definition of VC dimension which is based on probability below.

Definition10 (PVC-dimension): First, we suppose that as for all 2^N classifications of N samples, if the probability of one classification is smaller than the threshold t, it will not be taken into consideration. For classifications of N samples on the distribution f(x), one classification can be learned by hypotheses from set S only when the probability of training error $E_{in}(h)$ smaller than η. $p(\eta)$ is the probability of the inequality: $E_{in}(h) < \eta$ .The VC dimension of the hypothesis S is N when S cannot classify any N+1 samples.

The Inequality of PVC Bound

Now, we can apply new definitions mentioned before to the inequality of PVC bound on the base of original VC bound of Vapnik.

For the sample space X on distribution f(x), the size of sample set is N. S is the hypothesis set. The classifications whose probability are less than t will not be learned.

So, according to conditional probability equation, we have:

$$\begin{aligned} & p(|E_{in}(g) - E_{out}(g)| > \varepsilon, E_{in}(g) \leq \eta) \\ & = p(|E_{in}(g) - E_{out}(g)| > \varepsilon \mid E_{in}(g) \leq \eta) \times p(E_{in}(g) \leq \eta) \\ & = p(|E_{in}(g) - E_{out}(g)| > \varepsilon \mid E_{in}(g) \leq \eta) \times p(\eta) \end{aligned} \tag{11}$$

$p(\eta)$ is the probability which is confirmed by the error constant η. g is the hypothesis from hypothesis set S which minimizes $E_{in}(g)$. $E_{out}(h)$ is the ideal error of hypothesis h.

According to Vapnik, we can have the following inequality [4]:

$$\begin{aligned} & p(\sup_{\forall h \in S} |E_{in}(h) - E_{out}(h)| > \varepsilon \mid E_{in}(g) \leq \eta) \\ & \leq 2p(\sup_{\forall h \in S} |E_{in}(h) - E^{'}_{in}(h)| \geq \frac{\varepsilon}{2} \mid E_{in}(g) \leq \eta) \end{aligned} \tag{12}$$

$E_{in}(h)$ and $E^{'}_{in}(h)$ separately represents the training error of two training samples.

Next, we can proof the inequality below:

Theorem 3.1.

$$\begin{aligned} & p(\sup_{\forall h \in S} |E_{in}(h) - E^{'}_{in}(h)| \geq \frac{\varepsilon}{2} \mid E_{in}(g) \leq \eta) \\ & \leq 2m_s(2N) \times e^{-\frac{1}{8}\varepsilon^2 N} \end{aligned} \tag{13}$$

$m_s(2N)$ is the new growth function of hypothesis S.

Proof.
Define every 2N dimension sample:

$$X = \{x_1, x_2, ...x_N, x_1^{'}, x_2^{'}, ..., x_N^{'}\} \tag{14}$$

If we sort x1,x2,...xN,x1',x2',...,xN' into order, there are (2N)! different ways. Changing the sort of every sample X will construct another sample $X^{'}$ and it will not influence the proof of the inequality as a consequence of the symmetry.

Note the transformation matrix which turns X into $X^{'}$ as A, which means that:

$$X = \{x_1, x_2, ...x_N, x_1^{'}, x_2^{'}, ..., x_N^{'}\} \tag{15}$$

For every sample X , there will be (2N)! transformation matrices.
So we have this:

$$\begin{aligned} & p(\sup_{\forall h \in S} \left| E_{in}(h) - E_{in}^{'}(h) \right| \geq \frac{\varepsilon}{2} \mid E_{in}(g) \leq \eta) \\ & = \int_X \theta(\sup_{\forall h \in S} \left| E_{in}(h) - E_{in}^{'}(h) \right| - \frac{\varepsilon}{2}) dp \\ & = \int_X \frac{(\sum_{i=1}^{(2N)!} \theta_{A_i} (\sup_{\forall h \in S} \left| E_{in}(h) - E_{in}^{'}(h) \right| - \frac{\varepsilon}{2}))}{(2N)!} dp \end{aligned} \tag{16}$$

θ is a function:

$$\theta(x) = \begin{cases} 1, x \geq 0 \\ 0, x < 0 \end{cases} \tag{17}$$

θ_{A_i} is the situation when the sample X is transformed by matrix A_i .

Change the order of integration and summation, we can get:

$$\begin{aligned} & \sum_{i=1}^{(2N)!} \frac{(\int_X \theta_{A_i} (\sup_{\forall h \in S} \left| E_{in}(h) - E_{in}^{'}(h) \right| - \frac{\varepsilon}{2}) dp)}{(2N)!} \\ & = \frac{1}{(2N)!} \times \sum_{i=1}^{(2N)!} p_{A_i} (\sup_{\forall h \in S} \left| E_{in}(h) - E_{in}^{'}(h) \right| \geq \frac{\varepsilon}{2}) \end{aligned} \tag{18}$$

p_{A_i} represents the probability that sample in the function has been transformed by transformation matrix A_i .

Apparently:

$$\begin{aligned} & \frac{1}{(2N)!} \times \sum_{i=1}^{(2N)!} p_{A_i} (\sup_{\forall h \in S} \left| E_{in}(h) - E_{in}^{'}(h) \right| \geq \frac{\varepsilon}{2}) \\ & \leq \sum_{h \in S} \frac{1}{(2N)!} \times \sum_{i=1}^{(2N)!} p_{A_i} (\left| E_{in}(h) - E_{in}^{'}(h) \right| \geq \frac{\varepsilon}{2}) \end{aligned} \tag{19}$$

According to the theory by Vapnik [12] and the inequality of probability [13] we can get:

$$\frac{1}{(2N)!}\times\sum_{i=1}^{(2N)!}p_{A_i}(\left|E_{in}(h)-E_{in}^{'}(h)\right|\geq\frac{\varepsilon}{2})$$
$$=\sum_{\frac{2k-m}{N}\geq\frac{\varepsilon}{2}}\frac{C_m^k\times C_{2N-m}^{N-k}}{C_{2N}^N} \tag{20}$$
$$\leq 2e^{-\frac{1}{8}\times\varepsilon^2 N}$$

Also, the summation is $m_s(2N)$ according to the definition before. So this inequality is proved:

$$p(\sup_{\forall h\in S}\left|E_{in}(h)-E_{in}^{'}(h)\right|\geq\frac{\varepsilon}{2}\mid E_{in}(g)\leq\eta)$$
$$\leq 2m_s(2N)\times e^{-\frac{1}{8}\varepsilon^2 N} \tag{21}$$

According to the inequalities above, we can have:

$$p(\left|E_{in}(g)-E_{out}(g)\right|\mid E_{in}(g)\leq\eta)\leq 4m_s(2N)\times e^{-\frac{1}{8}\varepsilon^2 N} \tag{22}$$

And finally we can have:

$$p(\left|E_{in}(g)-E_{out}(g)\right|<\varepsilon, E_{in}(g)\leq\eta)$$
$$=p(\left|E_{in}(g)-E_{out}(g)\right|<\varepsilon\mid E_{in}(g)\leq\eta)\times p(\eta) \tag{23}$$
$$\leq 4m_s(2N)\times e^{-\frac{1}{8}\varepsilon^2 N}\times p(\eta)$$

Because taking probability into consideration, as for the same size of sample, the new hypothesis set is much smaller than the original one and will lead to the diminution of the growth function. This makes the bound more tight. In addition, the factor $p(\eta)$ also helps to make the bound smaller. So when the data set is confirmed, this VC bound inequality will be more stronger and allows the hypothesis set to classify bigger sample set without the loss of accuracy.

In another word, this means that when the bound and the hypothesis set are confirmed, we can classify bigger sample set and the result is just as good as before.

Conclusion

The new definition of VC dimension is of great meaning. When we design a classic classifier by using original VC dimension, we usually consider the training error and the range of confidence interval separately. Originally, these two factors always contradict to each other. We can get a target function of small training error with a huge VC dimension which means a huge range of confidence interval. It is always hard to find a balance before. However, the new definition of VC dimension takes the contribution of samples and probability into consideration. When we have some information of the sample set, this paper offers an approach to classify bigger sample set with the same hypothesis set. It means that our hypothesis set is more powerful. It will be so much useful because we can always have some information about data set in the real situation.

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